Experiment 3

(i) To calculate the energy and average power of given continuous-time and discretetime signals and classify them as energy signals, power signals, or neither based on their energy and power characteristics.

Expected Outcomes

- Calculate finite energy for an energy-type signal.
- Calculate average power for a periodic/power-type signal.
- Classify the signal based on the obtained values.
- Obtain numerical results consistent with theoretical expectations.

Software / Laboratory Environment

MATLAB

Formulae / Analysis

$E = \int_{-\infty}^{\infty} |x(t)|^2 \, dt$ for a continuous-time signal.

$P = \lim_{T\to\infty} (1/2T) \int_{-T}^{T} |x(t)|^2 \, dt$.

For $x(t) = \sin(2\pi f_0 t)$, average power is 1/2 for unit amplitude.

A signal with finite non-zero energy and zero average power is an energy signal; a finite nonzero average-power signal is a power signal.

Percentage deviation = |Practical − Theoretical| / |Theoretical| × 100.

Experimental Procedure

1. Define the time vector and generate the selected signals.
2. Compute the squared magnitude of each signal.
3. Numerically integrate over a sufficiently large interval for energy.
4. Compute average power over an interval containing several periods for the periodic signal.
5. Compare numerical results with analytical expectations.
6. Classify the signals and record observations.

Key Conditions & Threshold Values for Analysis

- The integration interval should be sufficiently long for decaying signals.
- For periodic signals, use an interval containing complete cycles where practical.
- Use adequate sampling resolution for numerical integration.
- Energy and power definitions must not be interchanged.
- For $x_1(t) = e^{(-t)}u(t)$, the theoretical energy is finite and equal to 0.5.

• For $x_2(t) = \sin(2\pi 5t)$, the theoretical average power is 0.5.

MATLAB Coding

```
% Experiment 3: Energy and Power Analysis
clc; clear; close all;
dt = 0.001;
t = 0:dt:23;
xE = exp(-t);
xP = sin(2*pi*5*t);
E_num = trapz(t,abs(xE).^2);
T0 = 1/5;
idx = t <= 2;
P_num = (1/(t(find(idx,1,'last'))-t(1))) * ...
 trapz(t(idx),abs(xP(idx)).^2);
figure;
subplot(2,1,1);
plot(t,xE); grid on;
title('Energy Signal e^{-t}u(t)');
xlabel('Time (s)'); ylabel('Amplitude');
subplot(2,1,2);
plot(t,xP); grid on;
title('Power Signal sin(2pi5t)');
xlabel('Time (s)'); ylabel('Amplitude');
fprintf('Numerical energy = %g\n',E_num);
fprintf('Numerical average power = %g\n',P_num);
```

Input and Output Graphs – Energy and Power Analysis

Observation

The numerical energy of $x_1(t) = e^{(-t)}\, u(t)$ is approximately 0.5000, and the numerical average power of $x_2(t) = \sin(2\pi 5t)$ is approximately 0.5000. The results agree with the theoretical values within the expected numerical accuracy.

Result

Energy and average power were calculated successfully, and the given signals were classified as energy and power signals based on their energy and power characteristics.

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage	Marks given	Marks obtained
Calculate the energy and average power of given signals	Energy & Power Calculation	Correctly calculates signal energy and average power using appropriate mathematical expressions and MATLAB implementation independently.	Performs calculations correctly with minor numerical or coding errors.	Performs basic calculations but requires guidance or has noticeable errors.	Unable to correctly calculate energy or power.	25	**2.5**	
Distinguish energy signals and power signals	Signal Classification	Correctly classifies signals as energy or power signals based on calculated values and theoretical conditions.	Correctly classifies signals with minor errors or limited justification.	Performs basic classification but requires assistance or provides incomplete justification.	Unable to classify the signal correctly.	25	**2.5**	
Implement numerical energy/power computation using MATLAB	MATLAB Implementation & Numerical Analysis	Develops accurate MATLAB code, obtains reliable numerical values and appropriately selects the integration/summation interval.	Code and numerical results are mostly correct with minor errors.	Implements the basic computation with assistance or noticeable numerical errors.	Unable to implement the required MATLAB computation.	25	**2.5**	
Compare analytical and simulated results	Theoretical Validation & Comparison	Accurately compares theoretical and simulated values, calculates deviation where applicable and explains discrepancies clearly.	Provides correct comparison with minor deviations and	Provides basic comparison but validation or interpretation is limited.	Unable to compare or validate the results.	25	**2.5**	